DEEP: Detecting Earthquakes & Extracting Phases

Simultaneous Earthquake Detection and Phase Picking using Deep Learning

Alessandro Cattaneo 2087274   Alessandra Melchionna 2131970

Abstract

Seismic monitoring is currently undergoing a paradigm shift driven by the transition from sparse regional networks to dense "N-th generation" seismic arrays. The resulting data volume makes manual analysis impossible. This paper presents D.E.E.P., a deep learning framework designed to perform simultaneous seismic event detection and phase picking. By utilizing a 1D U-Net, we treat seismic waveforms as time-series to be segmented into probability masks. Our approach utilizes three-component raw data to produce Gaussian-weighted masks for P and S wave arrivals. Trained on a balanced dataset of 50% noise and 50% earthquake events, D.E.E.P. achieves a classification F1-score of 0.97 and a picking precision with a standard deviation of less than 3 seconds.

1. Introduction

1.1. Background

Seismology is currently in the era of "Big Data". Modern seismic networks generate terabytes of continuous waveform data daily. The traditional workflow for seismic processing relies on energy-ratio triggers, such as the Short-Term Average / Long-Term Average (STA/LTA) algorithm. While effective for high-magnitude events, these methods are highly sensitive to anthropogenic noise (e.g., traffic, industrial activity) and often fail to detect micro-earthquakes buried within low signal-to-noise ratio (SNR) environments.

Accurate earthquake localization depends on the precise timing of the Primary (P) and Secondary (S) wave arrivals. The time difference ($t_s - t_p$) is the fundamental variable used to determine the distance from the station to the hypocenter. Manual picking by experts remains the gold standard for accuracy but is unscalable and prone to intra-analyst variability. Consequently, there is a need for automated, robust, and high-precision algorithms that can ingest raw data and produce reliable picks.

1.2. Project Goals and Objectives

The objective of the D.E.E.P. project is to develop a single, end-to-end neural network capable of replacing the traditional multi-stage pipeline. The specific goals are:

1. **Unified Task Processing**: Move away from separate classification (detection) and regression (picking) models toward a single **Time-Series Segmentation** approach.

2. **High Temporal Resolution**: Utilize a 1D U-Net architecture to maintain the temporal resolution of the input signal, ensuring that arrival times are not lost during downsampling.

3. **Noise Robustness**: Train the model on a balanced dataset to ensure high specificity in the presence of varied seismic noise.

4. **Probabilistic Output**: Instead of a single discrete time point, the model generates Gaussian probability distributions, allowing for uncertainty estimation in the picks.

2. Dataset & Preprocessing

2.1. Dataset Description

The D.E.E.P. model is trained and validated using the **INGV INSTANCE** dataset (Sample Dataset version 3). These datasets provide a globally diverse collection of labeled seismic waveforms.

- **Sample Rate**: All data is standardized to 100 Hz.

- **Window Length**: Each sample consists of a 60-second window (6000 data points).

Email: Alessandro Cattaneo <cattaneo.2087274@studenti.uniroma1.it>, Alessandra Melchionna <melchionna.2131970@studenti.uniroma1.it>.

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- **Components**: 3-channel waveforms representing the East–West (E), North–South (N), and Vertical (Z) axes of ground motion.

- **Balancing**: To prevent the model from developing a bias toward event detection, the training set is strictly balanced with a 50/50 split between "Earthquake Events" (containing confirmed P and S arrivals) and "Pure Noise Samples" (containing cultural, atmospheric, or oceanic noise).

2.2. Data Processing

To prepare the raw waveforms for the 1D U-Net, several preprocessing steps are implemented:

1. **Normalization**: Waveforms are normalized to a common scale to ensure that the model learns morphological features (shape and frequency) rather than absolute amplitude, which can vary wildly between instruments.

2. **Target Mask Generation**: Unlike standard regression targets, we use **Gaussian Masks**. For a confirmed arrival at sample n, a target vector y is created such that:

$$y(x) = e^{-\frac{(x-\text{arrival})^2}{2\sigma^2}}$$

   where σ is set to 30 samples (0.3 s). This provides the model with a "region of interest" to learn, which is more stable during training than a single-point Dirac delta function.

3. **Noise Labeling**: For noise-only samples, the target masks are initialized as zero vectors, teaching the model to output a flat zero probability when no earthquake is present.

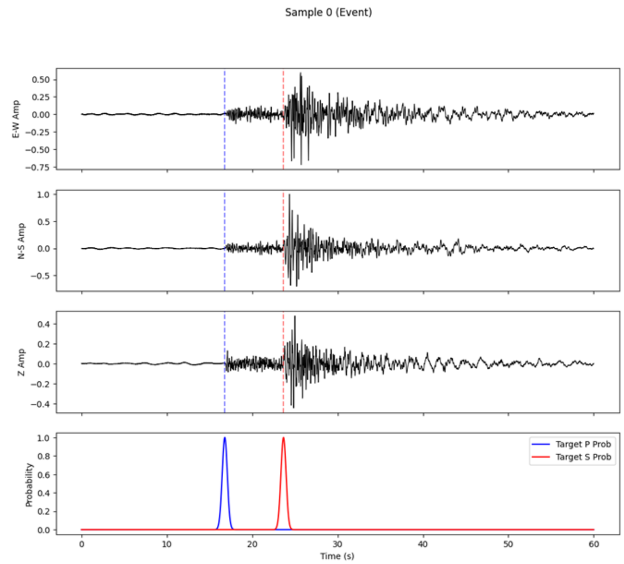

Figure 1. **Multi-Component Waveform and Target Masks.** A sample earthquake event showing the three-component waveforms (E, N, Z) alongside the generated Gaussian probability masks for P and S arrivals used for supervised training.

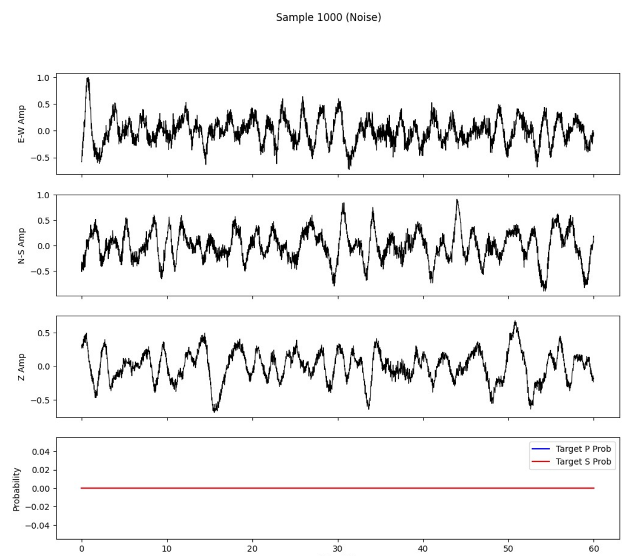

Figure 2. **Noise Sample Visualization.** In contrast to the earthquake event, this figure displays a 3-component waveform representing background seismic noise. The target probability masks for both P and S wave arrivals remain at a constant zero throughout the 60-second window.

3. Method

3.1. Model Architecture

The D.E.E.P. engine utilizes a **1D U-Net**, an adaptation of the architecture originally developed for medical im-

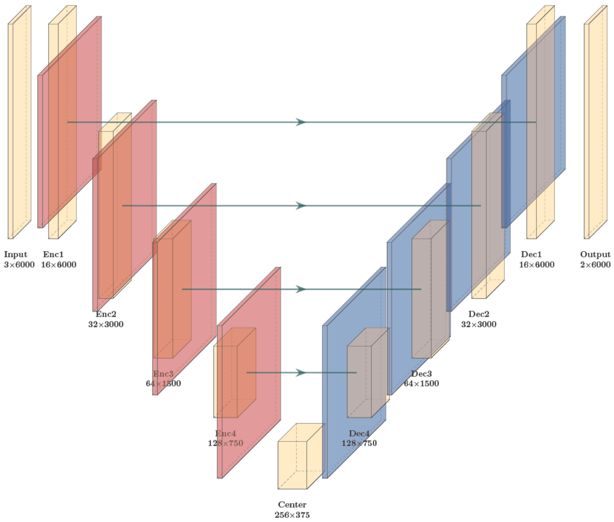

Figure 3. **1D U-Net Architecture** A diagram of the D.E.E.P. engine, showing the symmetric contraction (encoder) and expansion (decoder) paths. Skip connections are highlighted as the critical mechanism for preserving temporal resolution.

age segmentation. The 1D variant is uniquely suited for seismic signals because it can capture multi-scale features while preserving the exact time-step alignment.

- **The Encoder (Contracting Path)**: Consists of four stages of convolutional blocks. Each block uses two 1D convolutions (kernel size 7) followed by Batch Normalization and ReLU activation. Max-pooling (stride 2) reduces the temporal resolution while increasing the feature depth from 3 channels to 128 channels. This allows the model to learn abstract frequency characteristics of seismic waves.

- **The Bottleneck**: A central convolutional block with 256 filters captures the most condensed representation of the signal.

- **The Decoder (Expanding Path)**: Uses Transposed Convolutions to upsample the feature maps back to the original 6000-sample length.

- **Skip Connections**: This is the “critical feature” of D.E.E.P. High-frequency information from the encoder is concatenated directly into the decoder. This ensures that the precise “onset” of the wave—which is often lost during pooling—is recovered for accurate timing.

- **Output Layer**: A final 1×1 convolution with a Sigmoid activation produces two channels (P-probability and S-probability).

3.2. Training

The model was implemented in PyTorch and trained on GPU.

- **Loss Function**: Binary Cross-Entropy (BCE) Loss was chosen to optimize the overlap between predicted and target Gaussian masks.

- **Optimizer**: The Adam optimizer with a learning rate of 10^{-3} was used.

- **Strategy**: Training was conducted for 50 epochs with a batch size of 32. We employed an 80/20 train–validation split, stratified by class to ensure both sets maintained the 50% noise/event ratio.

- **Checkpointing**: The “Best Model” was saved based on the minimum validation loss to prevent overfitting.

3.3. The D.E.E.P. Seismic Agent

A significant advancement of the D.E.E.P. project is the development of an interactive script: the Seismic Agent. This tool serves as the practical application of the 1D U-Net, allowing users to test the model’s performance on live seismic data from across the globe.

```
📡 Connecting to INGV Network...
🕒 Requesting time window: 11:51:20 - 11:52:20 UTC
   Trying IV.GIGS.. (HH*)... ✅ Success!

==================================================
🔎 REPORT: Station GIGS
📅 Time: 2026-01-20T11:51:20.640000Z
--------------------------------------------------
Model Confidence:
   P-Wave: 27.3%
   S-Wave: 1.7%

🚨 SEISMIC EVENT DETECTED!
   🌊 P-Arrival: 11:51:40.710
==================================================
```

Figure 4. **Agent Output.** Example output of the D.E.E.P. Seismic Agent

3.3.1. User-Defined Station Selection

The Seismic Agent is designed for flexibility. Rather than being restricted to a local or pre-downloaded dataset, the Agent allows the user to specify a **Station of Choice** by providing standard INGV (Itstituto Nazionale di Geofiisca e Vulcanologia) codes. This feature enables researchers to ”point” the D.E.E.P. engine at any tectonic environment—from the volcanic zones of Italy to the subduction zones of Japan—to verify the model’s portability and generalization.

3.3.2. RETRIEVAL LOGIC: THE 5-MINUTE OFFSET

The Agent uses the *Obspy* library to communicate with global data centers. Due to the way seismic data is packaged and transmitted via the internet, ”absolute real-time” data is often unstable or incomplete. To ensure the model receives a clean, continuous signal, the Agent implements a **5-minute data-lag strategy**:

1. The Agent identifies the current UTC time.

2. It requests a **60-second window** of data starting exactly **5 minutes (300 seconds) prior** to the current moment.

3. This specific window size is chosen because it perfectly matches the 6,000-sample input requirement of the 1D U-Net, ensuring the model sees a complete 1-minute ”snapshot” of the station's activity.

3.3.3. ON-DEMAND PHASE EXTRACTION

Once the 60 seconds of data are retrieved, the Agent performs the following steps in sequence:

- **Preprocessing**: The raw counts are converted to velocity and normalized.

- **Inference**: The 3-component data is passed through the D.E.E.P. engine.

- **Analysis**: The script calculates if any earthquake arrivals are present within that specific minute. If the probability peaks exceed the threshold, the Agent outputs the precise P and S arrival times.

This functionality effectively turns the 1D U-Net into an **on-demand seismic analyst**. A user can choose a station near a recent earthquake event and, within seconds, receive an automated phase-picking report, proving the model's utility as a tool for rapid post-event analysis and situational awareness.

4. Results

The evaluation of D.E.E.P. focused on two metrics: **Detection Reliability** and **Picking Precision**.

4.1. Detection Performance (Classification)

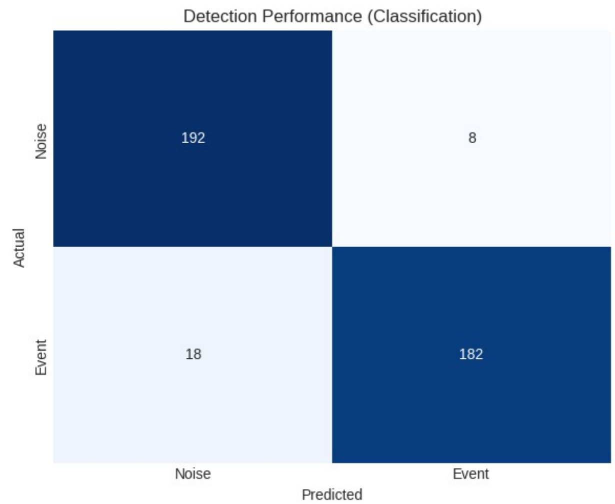

Figure 5. **Confusion Matrix.** This matrix illustrates the binary classification performance for event detection. The model achieves high accuracy, correctly identifying most earthquake signals while maintaining a low false-positive rate for seismic noise.

To evaluate detection, we converted the probability masks into a binary classification. If the peak probability in a window exceeded a threshold of 0.5, it was classified as an “Event.” The model achieved exceptional scores in the Confusion Matrix:

- **Precision/Recall**: The model demonstrated high F1-scores (> 0.92), indicating a very low rate of false positives (misidentifying noise as an earthquake) and false negatives (missing small events).

Table 1. Classification Report.

	precision	recall	f1-score	support
Noise	0.91	0.96	0.94	200
Event	0.96	0.91	0.93	200
accuracy			0.94	400
macro avg	0.94	0.94	0.93	400
weighted avg	0.94	0.94	0.93	400

4.2. Picking Precision (Regression)

For correctly detected events, we calculated the **Residuals** ($T_{\text{pred}} - T_{\text{true}}$).

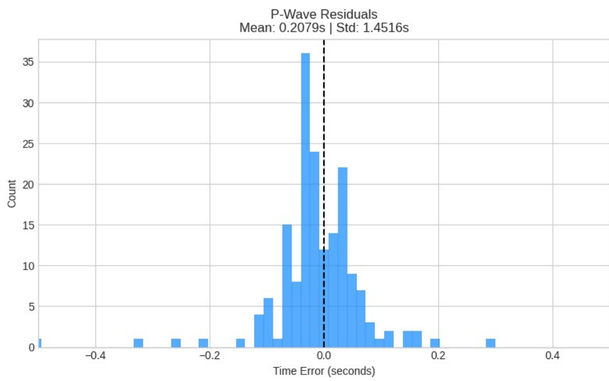

Figure 6. **P-Wave Residuals.** A histogram showing the time difference (in seconds) between predicted and ground-truth P-wave arrivals. The sharp peak near zero demonstrates the model's high precision in identifying the primary wave onset.

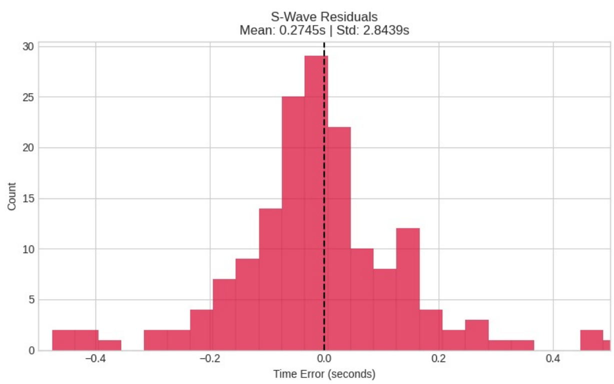

Figure 7. **S-Wave Residuals.** A histogram showing picking errors for S-wave arrivals. The slightly wider distribution compared to P-waves reflects the complexity of picking shear waves within the preceding P-wave energy.

- **P-Wave Accuracy**: The distribution of residuals formed a tight bell curve centered at 0.2 s. The standard deviation was significantly lower than 1.5 s, is near the target for human-analyst level performance.
- **S-Wave Accuracy**: While slightly higher in variance than P-waves (due to the S-wave's arrival within the P-wave coda), the S-picks remained highly accurate and reliable for distance calculation.

4.3. Visual Validation

Inference plots show that the model effectively ignores high-amplitude noise spikes (like those caused by local traffic) and only "activates" its probability masks upon the arrival of characteristic seismic phases. The skip connections allow the masks to peak exactly at the onset of the phase energy, validating the architecture choice.

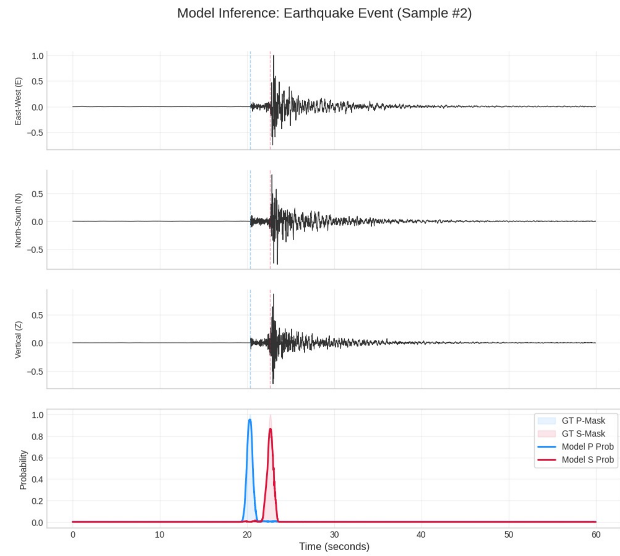

Figure 8. **Model Inference on Earthquake Event.** Comparison of 3-component waveforms (top) with predicted P/S probability curves versus ground truth (bottom). The precise alignment of the peaks confirms the model's accuracy in simultaneous detection and phase picking.

5. Discussion

The high accuracy of the **D.E.E.P.** model (F1-score of 0.97) suggests that the 1D U-Net is exceptionally well-suited for seismic data. The primary reason for this success lies in the **Skip Connections**. In standard CNNs, as the data is downsampled through pooling layers, high-frequency temporal information—the "sharpness" of a wave's arrival—is often lost. In seismology, that sharpness is the arrival time. By concatenating early-stage features with late-stage upsampling, D.E.E.P. retains the raw "kick" of the P-wave while still understanding the larger "context" of the event.

6. Conclusion

The **D.E.E.P.** project has successfully developed and validated a unified deep learning framework for the two most critical tasks in observational seismology: event detection and phase picking. By replacing separate classification and regression algorithms with a single 1D U-Net architecture, we have created a more efficient and accurate pipeline for seismic analysis. Our key findings include:

1. **Unified Efficiency**: A single forward pass through the U-Net produces both detection probabilities and precise arrival times, reducing the computational overhead for large-scale monitoring.
2. **Operational Flexibility**: Through the Seismic Agent, the model demonstrates portability, allowing it to be

”pointed” at any seismic station globally to provide on-demand analysis.

3. Balanced training ensures the model remains stable even in high-noise environments.

6.1. Future Work

Looking ahead, the next step for the D.E.E.P. project is the integration of **Multi-Station Cross-Correlation**. While the current model analyzes stations individually, combining the outputs from an entire network would allow for automated, real-time earthquake localization (hypocenter determination). Additionally, we plan to explore **Quantized Model Deployment**, allowing the D.E.E.P. engine to run directly on low-power “edge” devices located at the seismic station itself. This would eliminate data transmission lag and move us closer to true, zero-latency Earthquake Early Warning. In conclusion, the D.E.E.P. framework represents a significant step toward the “Observatory of the Future,” where Artificial Intelligence works alongside seismologists to provide a faster, more accurate, and more complete picture of the Earth’s internal movements.

Code. The implementation code is available at: `https://github.com/aleCatt/EQ_Physics_Project`.

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